



FACULTY OF COMPUTING

SECD2613 – 04

SYSTEM ANALYSIS AND DESIGN

2025/2026 – SEMESTER 2

PROJECT

**Phase 2- Information System Gathering and Requirement
(11%)**

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1.0 OVERVIEW OF THE PROJECT

In the high-pressure environment of university life, both students and lecturers are constantly juggling an overwhelming mental load. Between back-to-back lectures, intensive research and the endless cycle of assignments, staying afloat is a challenge. While many still rely on traditional paper planners or basic digital calendars, these methods often crumble under the weight of complex projects and high-stake deadlines. This lack of a cohesive system frequently leads to a fragmented workspace, mounting stress, and the inevitable "human error" that occurs when things get too messy.

The struggle is further complicated by "app fatigue." When essential information is scattered across various platforms, users spend more time switching between tabs than getting work done. This constant digital hopping is more than just a nuisance. It's a productivity drain that leaves students feeling burnt out and disorganized. Without a centralized hub, balancing academic rigor with extracurricular commitments becomes a losing game, often resulting in missed opportunities and a lingering sense of being overwhelmed.

Communication presents its own set of hurdles. While social media platforms like WhatsApp or Telegram are convenient for quick chats, they are in poor environments seeking help. Critical academic questions often get buried under a mountain of unrelated messages, and finding reliable answers later becomes a frustrating scavenger hunt.

This is exactly why EduHub was developed. It's designed to pull these moving parts into one intuitive, centralized home. By featuring a specialized Q&A forum where students can post specific hurdles and receive targeted answers, the system acts as a living knowledge base. Combined with comprehensive archives of past papers, recordings of physical lectures and smart tools like an AI-driven progress tracker, UTM Tutoring creates a supportive ecosystem that turns the chaos of university life into a structured path toward success.

2.0 PROBLEM STATEMENT

2.1 Fragmented Academic Tools

Normal teaching requires both lecturers and students to deal with several distinct platforms. There is no centralised platform for everything, hence the need for users to switch back and forth among different applications.

2.2 Unavailable Lecture Content

Those students who are unable to participate physically in class lessons due to sickness or any other personal issue have no way of accessing the lesson content, which would otherwise assist them during studies. Currently, there are no uniform guidelines which help in providing recorded lessons in real-time.

2.3 Limited Access to Academic Carry Mark

They do not get real-time and relevant data concerning their academic status and mark progression. It is common for students not to be aware that they are underperforming until it is too late for them to improve their academic situation during the semester.

2.4 Lack of a Centralised Academic Recourses

The past year papers, notes, and other academic material are not made available uniformly in personal computers and even through social circles. There is no institutional archive where such material can be stored and shared equally amongst all the students.

3.0 PROPOSED SOLUTION

3.1 Classroom Recording Archive

The EduHub application has a feature where lectures held in classrooms can be recorded and saved so that the students have access to the recordings. Students who cannot attend physically can watch the complete recording anytime. In this way, students will not face a disparity because of unavoidable reasons. There will be no need for lecturers to repeat themselves, as they will be able to focus their time on other issues.

3.2 Academic Forum

There will be an organized Q&A forum with subjects tagged for easy identification of questions raised and answers provided. Questions and answers will always remain indexed and searchable within the platform. Thus, there will be a living institutional memory that is growing rich every semester as far as knowledge acquisition is concerned. Students will get answers instantly while lecturers will be aware of what confuses their learners. EduHub also houses an extensively curated library of examination papers from previous years as well as academic sources, which are quality-controlled by the platform. All UTM users get the same access to the library material without favouritism or discrimination depending on their connections or background.

3.3 Performance Analysis Tracker

An AI study assistant analyses the learning pattern and performances of a user to come up with personalized recommendations. It identifies areas of knowledge gaps and tracks carry marks.

3.4 Study AI tools

By embedding AI tools directly into EduHub, the platform transforms from a passive storage space into an active, intelligent ecosystem that directly cures "app fatigue" and cognitive overload. AI acts as a centralized brain—instantly synthesizing hours of physical lecture recordings and vast archives of past papers into interactive, bite-sized study guides, while simultaneously powering the Q&A forum to route and answer complex academic hurdles automatically. Furthermore, by replacing rigid traditional calendars with a smart, predictive progress tracker, the system dynamically breaks down daunting deadlines into manageable daily tasks, effortlessly turning the chaotic, high-pressure mental load of university life into a structured, stress-free path to success.

3.5 Student Collaboration

This feature will allow users to connect or chat with seniors or lecturers in real time. This can help them gain more connections and possible opportunities related to their career in the future. Not only that, this can help students improve their networking skills as they get to display their expertise in the forum.

3.6 Study Guide Section

The most important part of studying is actually knowing how to study, which is something a lot of students do not know. This section contains multiple guides on how to study properly, authored by lecturers and contains segments taken from study guide books and articles. Topics include mindset management, diet, sleep management, study techniques and so on. This feature will also provide students with a built-in pomodoro timer which users can use. The timer starts when the user uploads their reading material and clicks start. After a set amount of time, for example 25 min, the timer will notify the user to take a break and continue studying again for another 25min. The cycle repeats until the user is satisfied, after which the system will provide a quiz for assessment and summary of their session.

4.0 INFORMATION GATHERING PROCESS

4.1 METHOD USED

During the information gathering process, we implemented two methods, which were questionnaires and Joint Application Design (JAD). The purpose of carrying out the information gathering process was to investigate the problems in the current e-learning system and identify the needs for improvement for EduHub, the proposed improved e-learning system.

The questionnaire consisted of three parts. The first part was about the demographic information of the respondents, the second part was related to the current e-learning system, and the third part was related to EduHub. Most of the questions were closed-ended questions, and the answer options were provided for the respondents to choose from. The target respondents were UTM students, and the number of respondents was 20. The questionnaire was created using Google Forms and distributed through WhatsApp. The purpose of this questionnaire was to collect general opinions from UTM students towards the current e-learning system and the proposed improvements for the system. The questionnaire was also designed to support both AS-IS analysis and TO-BE requirement identification by helping the team understand the current system limitations and identify users' needs for EduHub.

On 13 April 2026, we conducted a JAD session with a few selected users. Our selected users were Tan Yu Kai and Pon Xuen Lin. The session was held at BK4, N28, Faculty of Computing, and was conducted by Tinesh Gajendra, the session leader, and Lai Yan Cheng, who took on the role of scribe and observer simultaneously. The session started with Tinesh, the session leader, welcoming the users. After that, the session continued with a brief introduction about the proposed prototype of EduHub. Then, the users actively tried to strengthen their understanding of the prototype by asking questions about how the system works and other related matters. Finally, the session moved towards constructive criticism and feedback from the users.

The purpose of the session was to establish clear system and user interface requirements for EduHub. Through this session, our users could provide feedback on the prototype, while our team could better understand their requirements and expectations. This can help improve user satisfaction with the final prototype and increase their sense of ownership. Our team will also be able to understand the users' requirements more clearly and develop the system based on those requirements.

Through both methods, our team collected general user feedback and detailed user opinions. This helped our team identify the requirements for EduHub by analyzing the data collected. This can improve the satisfaction and experience of students when using the improved e-learning system. At the same time, it helps ensure that the proposed system has better usability.

4.2 SUMMARY FROM METHOD USED

4.2.1 Questionnaire

A total of 20 responses were collected from UTM students through the questionnaire. Most of the respondents were Year 1 students, which made up 75% of the total respondents, and most of them used the current e-learning system daily or several times a week, which made up 90% in total. This shows that the respondents were familiar with the current e-learning system and were suitable to provide feedback.

Based on the responses, the current e-learning system is mainly used for downloading lecture materials, submitting assignments, downloading assignment questions, and accessing lecture recordings. Specifically, 90% of respondents used it to download lecture materials, 85% used it to submit assignments, 80% used it to download assignment questions, and 35% used it to watch lecture recordings. However, they think the current e-learning system still lacks academic support features. This causes them to try other ways to get academic help, such as through AI tools, friends, or lecturers. There will be a problem when using ways outside the e-learning system to get academic help, as 55% of respondents stated that answers may be unreliable, while 25% stated that responses from others may be slow or unavailable.

The questionnaire also showed several limitations in the current system. The first limitation is that past year questions are available for some subjects only, which was selected by 70% of respondents. This has made students unable to practise and strengthen their knowledge before the final examination. The second limitation is that carry marks are usually received near the final examination, as 60% of respondents stated that they receive their carry marks before the final examination. This makes it difficult for students to make their study plan according to their target grades and carry marks. There will be a problem where students may ignore lower-mark subjects because they do not know their carry marks early enough. In addition, respondents also stated that inconsistent organization of materials and slow system performance affected their experience when using the current e-learning system, with 50% selecting inconsistent organization of materials and 30% selecting slow or unstable system performance. In the aspect

of non-functional requirements, 35% of respondents agreed that only enrolled students can access course materials and recordings, while 25% agreed that only lecturers or admins can upload official materials and release marks. This helps to make sure that the e-learning system is reliable and secure.

For the proposed improvements, students showed interest in several features, including student-uploaded recordings, academic forum, AI study assistant, grade prediction, Pomodoro timer, and senior connection. Among these, senior connection, grade planning support, academic forum, and student-uploaded recordings were seen as useful improvements for EduHub. Senior connection is useful as 60% of respondents rated it as very useful, and students need advice on courses and projects or assignments so that they have a clear understanding about the courses they study and the projects they do. For the grade planning feature, it is useful because 65% of respondents stated that they would use it, and 55% of respondents selected the feature that lets students simulate possible future marks and view the predicted final grade. This helps students to know which part they need to focus on the most. For the academic forum, 75% of respondents stated that they would use it, and students think that the useful functions in it are sharing solutions and explanations, selected by 45% of respondents, and sharing past year questions, selected by 40% of respondents. This is because it will make it easier for them to get practice resources and explanations from others when they have questions related to certain subjects. For student-uploaded recordings, 65% of respondents stated that they would use this feature. It is useful for students who are unable to attend class for certain reasons, as they can catch up with the progress by watching the recordings. However, 55% of respondents think that recordings must be verified and approved by lecturers before being uploaded to prevent irrelevant recordings.

Overall, the questionnaire results show that the current e-learning system supports basic academic activities, but it does not provide enough academic discussion, grade planning, resource sharing, and student support features. Therefore, the findings support the need to improve the current e-learning system into EduHub.

Questionnaire Link : [EduHub](#)

4.2.2 JOINT APPLICATION DESIGN(JAD)

From the JAD session, the selected users gave three main opinions regarding the proposed system. The first suggestion was to add a report button feature in the forum. This feature can help maintain proper forum activity by allowing users to report inappropriate, irrelevant, or non-academic posts. As a result, the forum can remain clean, comfortable, and focused on academic purposes only.

The second suggestion was to add a “Request Lecturer Clarification” button in the forum. This feature is useful when a solution or answer contains ambiguity. By clicking the button, the lecturer will be notified and can provide clarification or explanation when available. This can help improve the accuracy and reliability of the answers in the forum.

The third suggestion was to add a reward system in the student forum to motivate students to answer other students’ questions. For example, students may receive points, badges, vouchers, or other rewards for providing useful answers. This can encourage more students to participate actively in the forum and make them feel that their contribution is valued.

Overall, the JAD session showed that users expect the forum feature to have proper moderation, reliable academic answers, and motivation mechanisms to encourage active student participation.

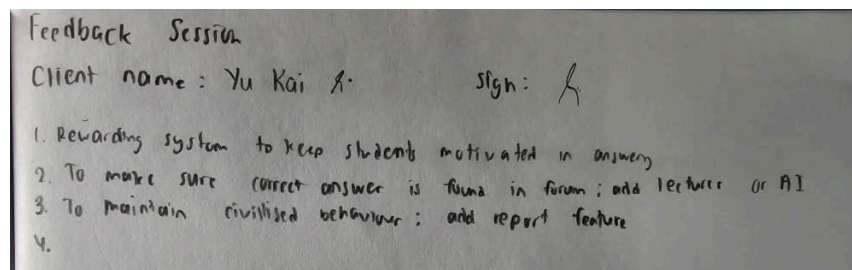


Figure 4.2.1a shows proof of feedback JAD Session

*Figure 4.2.2 proof of
involvement in
JAD Session*



5.0 REQUIREMENT ANALYSIS

5.1 Current business process (scenarios, workflow)

The questionnaire confirmed that 90% of respondents use the current e-learning system daily or several times a week, primarily for downloading lecture materials (90%), submitting assignments (85%), downloading assignment questions (80%), and accessing lecture recordings (35%). The following sub-sections describe the AS-IS workflow for each of the six EduHub module areas, grounded in these questionnaire findings and JAD feedback.

MODULE 1. SMART CLASSROOM RECORDING SYSTEM

AS-IS Scenario: Accessing Missed or Recorded Lecture Content

Only 35% of respondents use the current elearning system to watch lecture recordings, indicating that recordings are either unavailable or inconsistently uploaded. When a student misses a class or needs to review content:

#	Actor	Action	Current Tool	Problem
1.	Student	Misses a lecture or needs to re-watch content	UTM e-learning Portal	No standardised recording system; recordings not always available
2.	Student	Checks e-learning portal for uploaded slides or recordings	UTM e-learning Portal	Slides only. No audio/visual context; recording available for some subjects only.
3.	Student	Messages or email lecturer for re-explanation	WhatsApp/Email	Lecturer time wasted; inconsistent response; no scalable solution
4.	Lecturer	Responds individually or directs student to slides	WhatsApp/Email	Same explanation repeated many times; no share record

Questionnaire finding: 65% of respondents stated they would use a student-uploaded recording feature, and 55% indicated that recordings must be verified by lecturers before being uploaded. This confirms that students want accessible recordings but with a quality control mechanism.

MODULE 2. ACADEMIC FORUM AND DISCUSSION BOARD

AS-IS Scenario: Seeking Academic Help Outside Class Hours

The questionnaire showed that 55% of respondents find answers from external sources (peers, AI tools, lecturers) to be unreliable, and 25% experience slow or unavailable responses. This reflects the current state of academic help-seeking:

#	Actor	Action	Current Tool	Problem
1.	Student	Encounters difficulty with course material or question during self-study	Lecture notes	No immediate, structured channel for academic help
2.	Student	Posts question in WhatsApp or Telegram class group	WhatsApp/Telegram	Question buried under unrelated messages
3.	Peer	Responds if available	WhatsApp/Telegram	55% of respondents say answers from others may be unreliable
4.	Student	Searches AI tools or Google for answers	ChatGPT/Google	Generic answers not aligned to UTM syllabus
5.	Student	Contacts lecturer directly if still unanswered	WhatsApp/Email	25% say responses from lecturers/peers are slow or unavailable
6.	All users	Discussion is permanently lost after chat scroll	None	No archive; same question repeated every semester

Questionnaire finding : 75% of respondents stated they would use the academic forum feature, 45% want solution and explanation sharing, and 40% want past year question sharing in the forum.

JAD finding: Users suggested adding a report button for inappropriate posts, a ‘Request Lecturer Clarification’ button for ambiguous answers, and a reward system(points, badges) to motivate student participation.

MODULE 3. CARRY MARK AND PERFORMANCE ANALYSTIC

AS-IS Scenario: Tracking Academic Carry Marks

60% of respondents reported receiving their carry marks only before the final examination leaving them unable to adjust their study plan in time. The current workflow is:

#	Actor	Action	Current Tool	Problem
1.	Student	Completes quiz, assignment, or lab work	Physical submission/ UTM elearning	Mark not immediately visible after submission
2.	Lecturer	Records marks manually in spreadsheets	Excel	Delayed, inconsistent update schedule across lecturers
3.	Student	Checks UTM e-learning when marks are eventually published	UTM e-learning	Only final aggregate shown; no breakdown by assessment component
4.	Student	Manually tallies marks from returned physical papers	Calculator/Paper	Error-prone; incomplete if papers not returned
5.	Student	Realises underperformance only near semester end	Final result slip	Too late to improve; 60% receive marks only before final exam

Questionnaire finding : 65% of respondents stated they would use a grade planning feature, and 55% specifically want the ability to simulate future marks and view their predicted final grade. This highlights demand for a real-time analytics dashboard with predictive capabilities.

MODULE 4. AI STUDY ASSISTANT

AS-IS Scenario: Preparing for Exams and Assignments

Respondents indicated they turn to AI tools, friends, and lecturers for academic help when the e-learning system does not provide sufficient support. The current study preparation workflow is:

#	Actor	Action	Current Tool	Problem
1.	Student	Identifies topic to study from lecturer slides	PDF slides	Slides lack depth; no interactivity or personalisation
2.	Student	Uses external AI tools for explanation	ChatGPT/ Claude	Not aligned to UTM syllabus; 55% find such answers unreliable
3.	Student	Asks peers in group for chat for tips or summaries	WhatsApp/ Telegram	Peer knowledge varies; not always available or unreliable
4.	Student	Attempts past year questions alone with no feedback	Printed PYQ/ PDF	70% say PYQs are only available for some subjects; no automated feedback
5.	Student	Visits lecturer during consultation hours	Face-to-face/ Email	Limited slots; not available 24/7; same explanation repeated

Questionnaire finding: Students explicitly identify AI study assistants as a desired improvement. The combination of unreliable external AI tools and limited lecturer availability confirms the need for a course-aligned, integrated AI assistant within EduHub.

MODULE 5. STUDENT COLLABORATION FEATURES

AS-IS Scenario: Seeking Advice from Seniors

60% of respondents rated the senior connection feature as very useful, indicating strong demand for structured peer mentoring. Currently there is a few formal channel for students to connect with seniors within the academic platform but not officially :

#	Actor	Action	Current Tool	Problem
1.	Junior Student	Needs advice on courses, projects, or assignments	Personal networks/ WhatsApp	Access depends entirely on existing personal connections
2.	Junior Student	Searches social media or informal channels for seniors	Instagram/ Facebook/ Telegram	Unstructured; no academic context; may get outdated advice
3.	Senior Student	Responds if personally known; otherwise unavailable	WhatsApp	No platform incentive to share knowledge; knowledge not archived

Questionnaire finding: Senior connection was the most high rated proposed feature (60% ‘very useful’). Students cited needing advice on courses and projects as the primary reason, confirming the need for a structured in-platform mentoring and collaboration channel.

MODULE 6. STUDY TIPS/GUIDE SECTION

AS-IS Scenario: Studying Without Guidance or Structured Resources

Currently, there is no formal channel within the UTM academic platform for students to access study tips, structured guides, or past year questions. Students rely entirely on informal methods to find study resources:

#	Actor	Action	Current Tool	Problem
1.	Student	Searches for past year papers online	Google	70% say PYQs only available for some subjects; reliability unknown
1.	Student	Study the course material without any guide	PDF slides	Having a mental block on how to study
2.	Student	Ask other Peers on how to study	WhatsApp	The tips given are not suitable for them self
3.	Student	Uses a phone timer or external app for study sessions	Phone timer	No integration with course material; no post-session feedback
4.	Lecturer	Occasionally uploads study guides to e-learning portal	UTM e-Learning	Inconsistent; only benefits students who check promptly; no structured archive
6.	All users	Resources are lost at end of semester	None	No institutional archive; each new cohort start from scratch

Questionnaire finding: 70% of respondents said past year questions are only available for some subjects. 40% identified PYQ sharing as the most useful forum feature. Students also expressed interest in a Pomodoro study tool integrated with their uploaded material.

5.2 Functional Requirement (input, process and output)

The following functional requirements are derived directly from the AS-IS analysis (section 5.1), questionnaire findings, and JAD session feedback. Each requirement is presented with its triggering input, the system process, and the expected output.

5.2.1 Student Functions

FR NO.	Current Function	Input	Process	Output
FR-01	A student shall download lecture materials from the UTM e-learning portal	Student chose the material they needed	Portal retrieves uploaded lecture files	PDF slides downloaded successfully
FR-02	A student shall download assignment questions from the e-learning portal	Student chose the assignment question	Portal retrieves assignment question files uploaded by the lecture	Assignment file downloaded
FR-03	A student shall submit completed assignments through the UTM e-learning portal.	Student uploads a completed assignment file on the portal before the submission	Portal receives the uploaded file and stores it linked to the subject	Submission recorded and confirmation shown
FR-04	A student shall access available lecture recordings uploaded on the UTM e-learning portal.	Student looks for any uploaded video recordings	Portal retrieves any video files uploaded by the lecturer and streams or provides a download link	Recording accessed if available
FR-05	A student shall view published carry marks and assessment results on the UTM e-learning portal.	Student navigates to the Grades or Results section of the portal	Portal retrieves mark entries published by the lecturer and displays them to the student	Marks displayed when published

5.2.2 Lecturer Functions

FR NO.	Current Function	Input	Process	Output
FR-06	A lecturer shall upload lecture materials to the UTM e-learning portal for student access.	Lecturer prepares lecture slides or notes in PDF or document format	Lecturer logs into the portal and uploads the file to the relevant subject page manually	File published and downloadable by enrolled students — but upload is voluntary and inconsistent
FR-07	A lecturer shall upload assignment questions to the portal and set a submission deadline.	Lecturer prepares assignment question file and sets deadline date	Lecturer uploads the assignment file to the portal and configures the submission deadline	Assignment visible to students with deadline shown
FR-08	A lecturer shall download and review student assignment submissions from the portal.	Students have submitted assignments through the portal	Lecturer logs into portal, navigates to the submission list, and downloads individual or batch student files	Submissions downloaded for marking
FR-09	A lecturer shall upload lecture recordings to the portal on a voluntary basis.	Lecturer has a recorded video of a lecture session	Lecturer uploads video file manually to the subject page on the portal	Recording published for student access — upload is entirely voluntary; no standard enforced across subjects; only 35% of students access recordings, suggesting most lecturers do not upload them
FR-10	A lecturer shall publish student marks on the portal after assessment completion.	Lecturer has completed marking of quizzes, assignments, or examinations	Lecturer manually enters or uploads marks into the portal's grade book for each student	Marks published and visible to the respective student, entry is manual and delayed; 60% of students receive marks only near the final examination; no automated alert when marks are released

5.3 Non-functional Requirement (performance and control)

The following describes the quality issues of the Current System. We will see the current informal tools perform in terms of performance and control.

5.3.1 Performance

NFR NO.	Category	Current Weakness	Evidence from Questionnaire/JAD
NFR-01	Response Time	The current UTM e-learning portal shall experience slow and unstable performance, especially during peak periods such as assignment submission deadlines or exam season.	30% of respondents cited slow or unstable system performance as a key limitation of the current e-learning system.
NFR-02	Reliability	The current system shall have no guaranteed uptime or service continuity for academic purposes. WhatsApp and Telegram, the primary academic communication tools, are not purpose-built for academic issues. Notifications are mixed with personal messages, making academic content hard to locate quickly	90% of respondents use the e-learning system daily, yet still rely on informal tools
NFR-03	Response speed for academic help	When a student posts a question in a group chat, there is no guaranteed response time. Responses depend entirely on peer availability and willingness	25% of respondents stated that responses from peers or lecturers are slow or unavailable

5.3.2 Control

NFR NO.	Category	Current Weakness
NFR-04	Authentication	The current portal shall require users to authenticate using their official UTM student or staff credentials before accessing any content. Unauthenticated users cannot access course materials, submissions, or grade information. This control is enforced through the UTM single sign-on system.
NFR-05	Data Privacy	The current portal shall restrict each student's marks and personal academic data to their own account view only. A student cannot view another student's grades or submissions through the portal
NFR-06	Assignment Submission Integrity	The current portal shall record a timestamp and student ID for every assignment submission. Each submission is linked to the student's account, preventing unauthorized or anonymous submissions. However, the portal does not enforce file type or size restrictions consistently across all subjects.
NFR-07	Student Access	The current portal shall restrict access to course materials and recordings to enrolled students only. A student must be officially registered in a subject to view its content. 35% of respondents confirmed agreement that only enrolled students should be able to access course materials, indicating this control is recognised by users as important.
NFR-08	Lecturer Upload Rights	Only lecturers and administrators shall be able to upload official lecture materials, assignment questions, and marks to the portal. Students do not have upload rights for academic content. 25% of respondents agreed that only lecturers or admins should upload official materials and release marks, confirming this constraint exists in the current system.

5.4 Logical DFD AS-IS system (Context Diagram, Diagram 0, Child)

5.4.1 Context Diagram of the AS-IS system

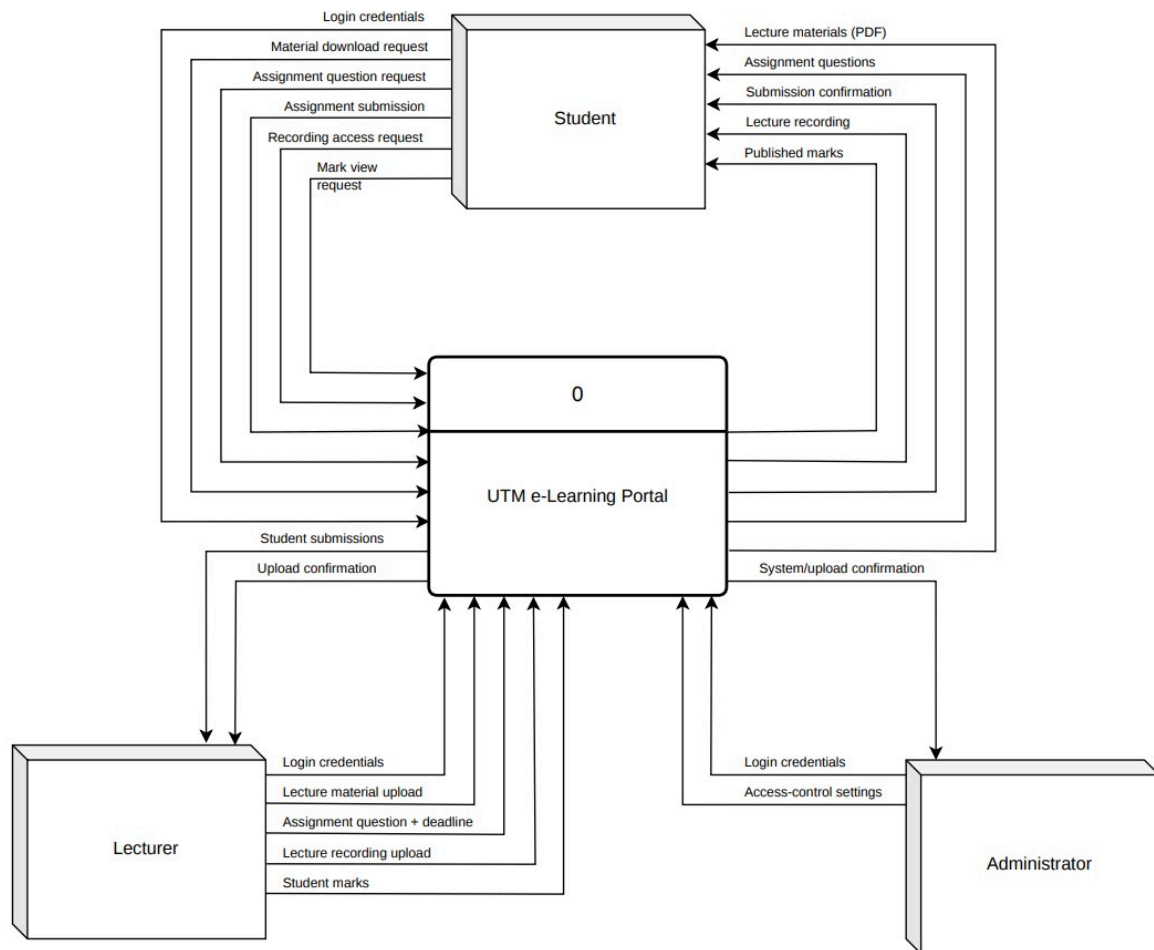


Figure 5.4.1 shows Context Diagram

5.4.2 Diagram 0 of the AS-IS system

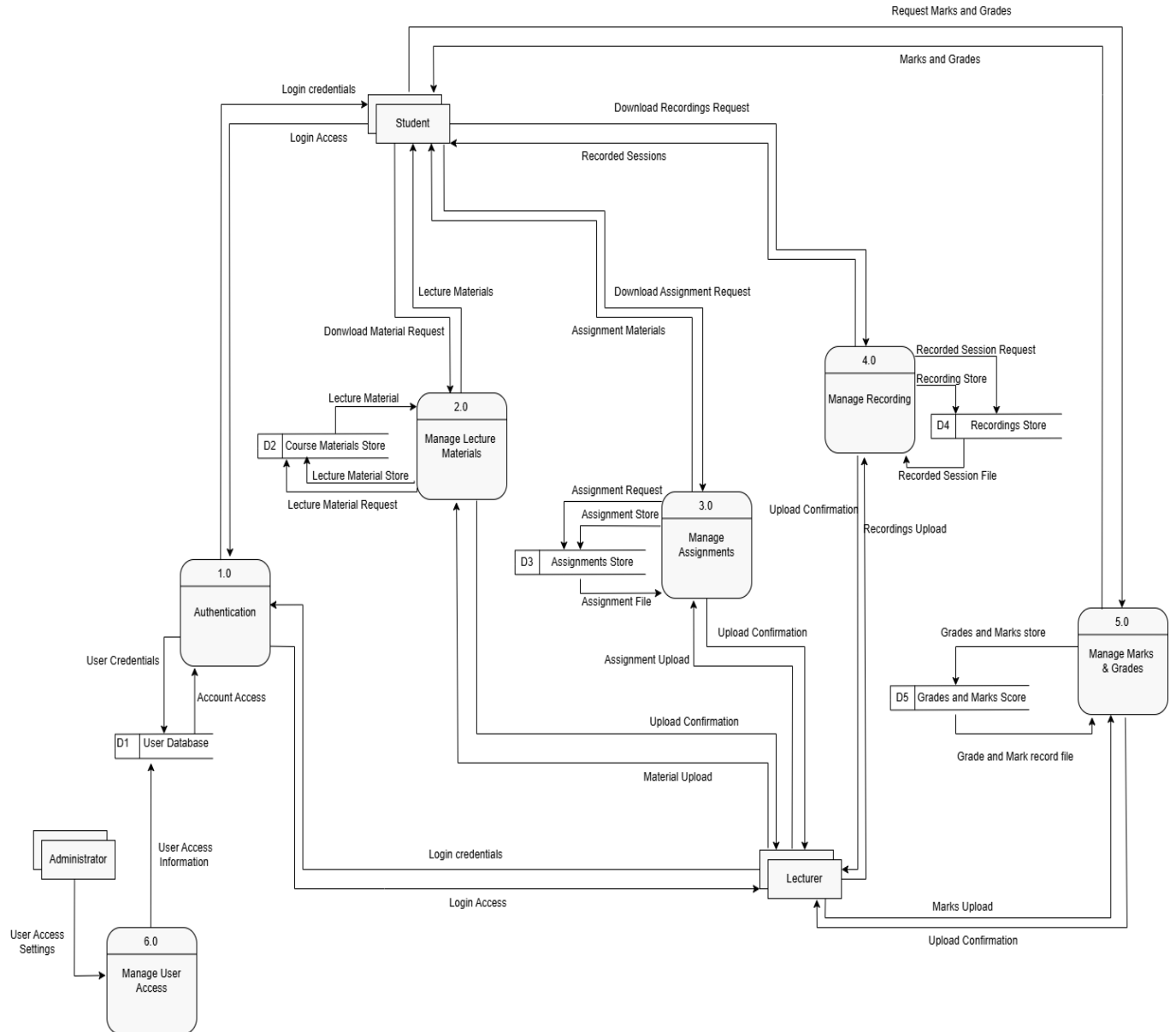


Figure 5.4.2 shows Diagram 0

Note: Only '1.0 Authentication' and '6.0 Manage User Access' will require child diagrams. The rest of the subprocesses are primitive processes already.

5.4.3 Child Diagram

Child Diagram for 1.0 Authentication

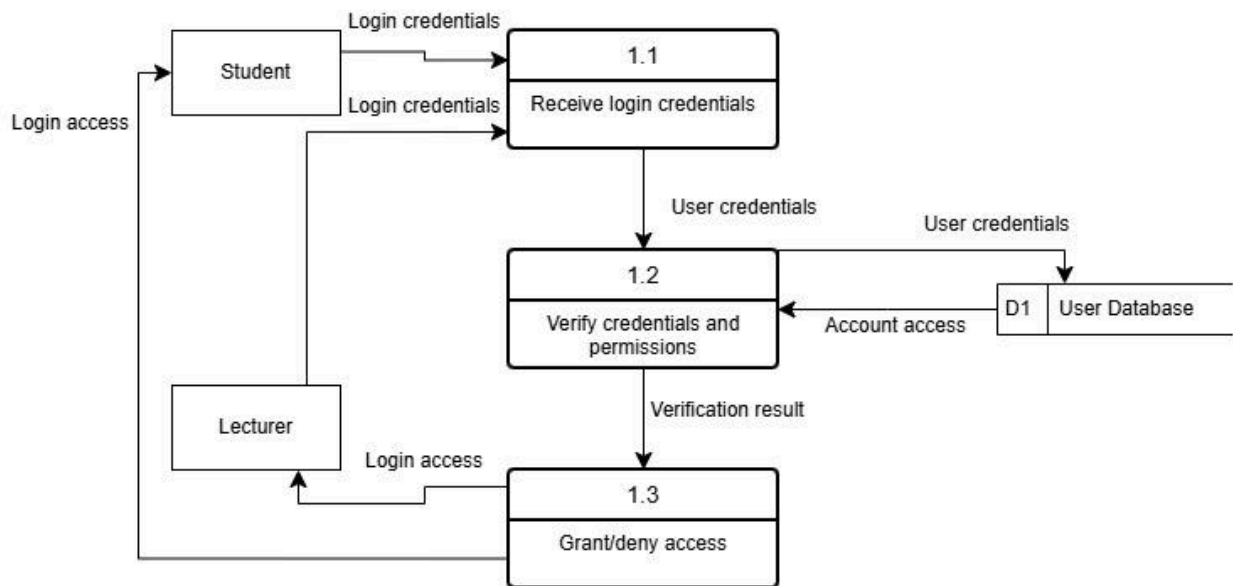


Figure 5.4.3a shows Child Diagram for 1.0 Authentication

Child Diagram for 6.0 Manage User Access

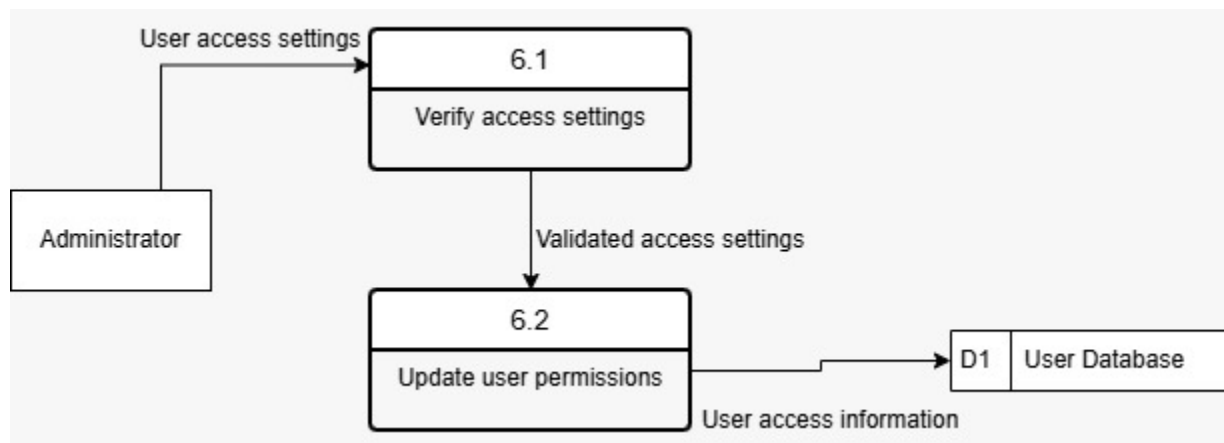


Figure 5.4.3b shows Child Diagram for 6.0 Manage User Access

6.0 Summary of Requirement Analysis

The requirement analysis for EduHub was carried out based on the data collected through two information gathering methods, which were a questionnaire involving 20 UTM students and a Joint Application Design (JAD) session conducted with two selected users. The findings from both methods were used to support the AS-IS analysis of the current e-learning system and the TO-BE requirement identification for EduHub.

The AS-IS analysis showed that the current e-learning system only supports basic academic activities such as downloading lecture materials, submitting assignments, and accessing limited recordings. It does not provide sufficient support for academic discussion, grade tracking, resource sharing, and student collaboration. As a result, students rely on informal tools such as WhatsApp, Telegram, and external AI tools, which lead to unreliable answers, scattered resources, and the loss of academic knowledge at the end of each semester. These weaknesses were observed consistently across all six module areas, namely the Smart Classroom Recording System, Academic Forum, Carry Mark and Performance Analytics, AI Study Assistant, Student Collaboration, and Study Tips/Guide Section.

Based on the AS-IS findings, a total of 10 functional requirements (FR-01 to FR-10) were identified and presented in the form of input, process, and output. These requirements describe how the current functions are carried out using informal methods and highlight the limitations that EduHub aims to overcome. In addition, eight non-functional requirements (NFR-01 to NFR-08) were identified, covering performance aspects such as response time, reliability, and response speed, as well as control aspects such as authentication, data privacy, assignment submission integrity, student access, and lecturer upload rights. Some requirements were supported with evidence gathered from the questionnaire and JAD session to ensure that the analysis reflects the actual function of the system.

The findings from the questionnaire further strengthened the identified requirements. Features such as the academic forum (75%), grade planning (65%), student-uploaded recordings (65%), and senior connection (60%) received strong interest from the respondents. The JAD session also contributed additional requirements, including a

report button for inappropriate posts, a "Request Lecturer Clarification" button, and a reward system to encourage active student participation in the forum.

Overall, the requirement analysis confirmed that the current e-learning system is unable to fully support the academic needs of students. The identified functional and non-functional requirements provide a clear and structured foundation for the design of EduHub, ensuring that the proposed system directly addresses the weaknesses of the current system and improves the overall academic experience of UTM students.